RAZAVI ELECTRONICS

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NAME: Sohail Ansari STD.: SEC.: ROLL NO.: SUB.: Analog Circuit ~~Signals & Systems~~.

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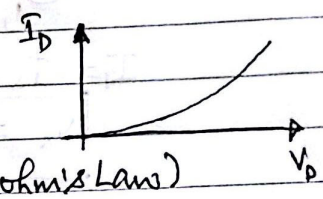
LECTURE : 06

MODELING OF P-N JUNCTION

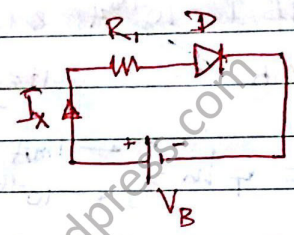
- **Exponential Model**

$$I_D = I_S \exp\left(\frac{V_x}{V_T}\right)$$



(does not satisfy ohm's Law)

Example



Assume Diode is Forward Biased.

$$V_B = I_x R_1 + V_D$$

$$V_B = I_x R_1 + V_T \ln \frac{I_x}{I_s}$$

To Solve I_x → Use Iterative Method.

If $I_x > 0$ then our assumption is Correct

If $I_x < 0$ then Diode is in off state.

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Lecture No : 06

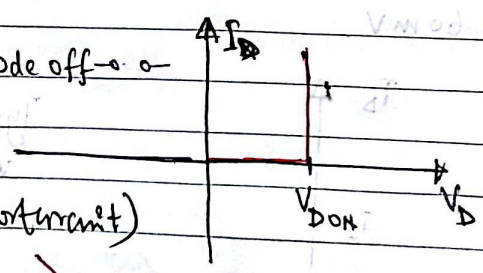
- **Constant voltage Model**

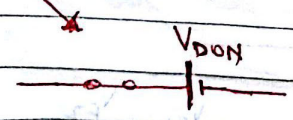
If $V_D < V_{DON}$ Diode off

If $V_D > V_{DON}$

Diode turns ON

(shortcircuit)

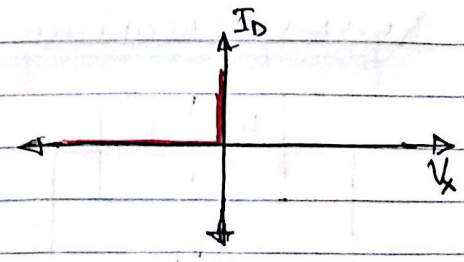




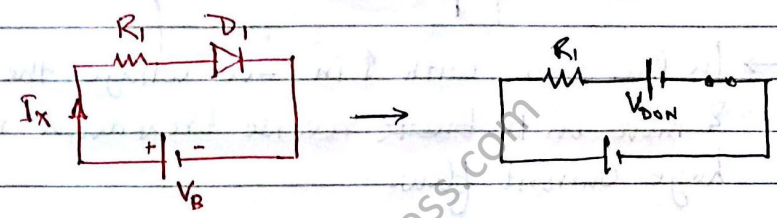
$V_{DON} = (700-800)mV$

Ideal Diode Model :

If $V_x > 0$ Diode ON
If $V_x < 0$ Diode OFF



Example

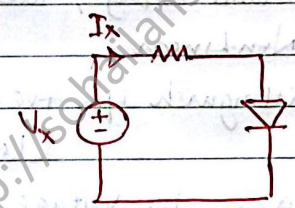


$V_B = I_x R_1 + V_{D(ON)}$

$$\boxed{I_x = \frac{V_B - V_{DON}}{R}}$$

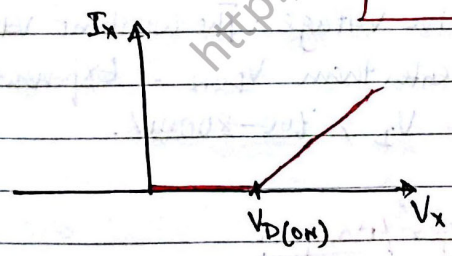
- Easy to Find
- Accuracy is less than exponential Model.

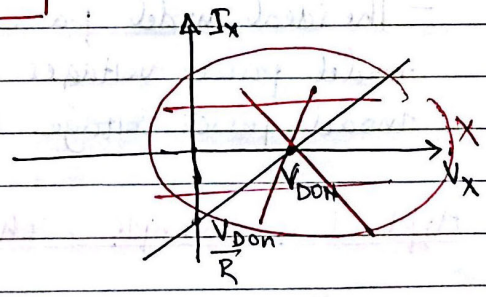
Quiz



Plot I_x Vs V_x when Diode ON

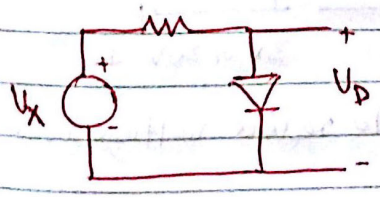
$I_x = \frac{V_x}{R} - \frac{V_{D\,on}}{R}$

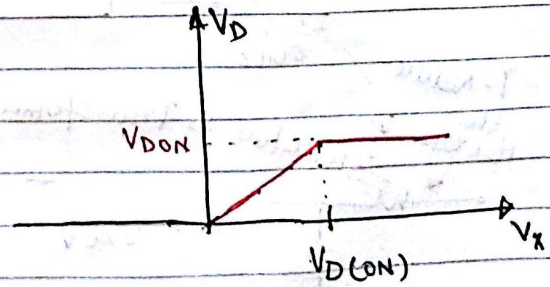




Example

Plot V_D Vs V_x





$V_x = I_x R_1 + V_D$

$V_D = V_x - I_x R_1 = V_{DON}$
(when D → F.B.)

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